



Direct detection of alpha satellite DNA with single-base resolution by using abasic Peptide Nucleic Acids and Fluorescent *in situ* Hybridization

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ABSTRACT

The detection of repetitive sequences with single-base resolution is becoming increasingly important aiming to understand the biological implications of genomic variation in these sequences. However, there is a lack of techniques to experimentally validate sequencing data from repetitive sequences obtained by Next-Generation Sequencing methods, especially in the case of Single-Nucleotide Variations (SNVs). That is one of the reasons why repetitive sequences have been poorly studied and excluded from most genomic studies. Therefore, in addition to sequencing data, there is an urgent need for efficient validation methods of genomic variation in these sequences. Herein we report the development of chemFISH, an alternative method for the detection of SNVs in repetitive sequences. ChemFISH is an innovative method based on dynamic chemistry labelling and abasic Peptide Nucleic Acid (PNA) probes to detect *in situ* the α -satellite DNA, organized in tandem repeats, with single-base resolution in a direct and rapid reaction. With this approach, we detected by microscopy the α -satellite DNA in a variety of human cell lines, we quantified the detection showing a low coefficient of variation among samples (13.16%–25.33%) and we detected single-base specificity with high sensitivity (82.41%–88.82%). These results indicate that chemFISH can serve as a rapid method to validate previously detected SNVs in sequencing data, as well as to find novel SNVs in repetitive sequences. Furthermore, the versatile chemistry behind chemFISH can lead to develop novel molecular assays for the *in situ* detection of nucleic acids.

1. Introduction

The detection of nucleic acids preserving their spatial location in each individual cell or tissue allows to establish precise molecular profiles. Most of the techniques for such purpose are based on *in situ* hybridization (ISH), and, currently, some of these techniques are being used to sequence *in situ* the detected nucleic acids (Ke et al., 2013; Payne et al., 2020; Asp et al., 2020). That results in a complete picture to understand the biological mechanisms involving these nucleic acids. In the

field of ISH techniques, Peptide Nucleic Acids (PNAs) have shown their potential as efficient probes. These PNAs are analogues to DNA, where the sugar-phosphate backbone is replaced by a peptide backbone formed by units of N-(1-aminoethyl)-glycine (Saarbach et al., 2019; Gupta et al., 2017). Such modification results in probes with absence of charges, thus displaying high affinity to their DNA targets (without electrostatic repulsion with the negatively-charged sugar phosphate backbone), forming more stable PNA-DNA duplexes than the natural DNA or RNA duplexes (Nielsen et al., 1991). Previous studies showed that PNA probes

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require a lower free energy (ΔG°) to bind to their DNA target sequence, which means that the formation of the PNA:DNA heteroduplex has a lower entropic cost than the formation of the DNA:DNA homoduplex (Ratilaan THolmén et al., 2005). This high affinity is reflected in the high dissociation temperature (T_m) that PNA:DNA heteroduplexes present, which can be around 1 °C higher for each base pair that hybridizes compared to the T_m of the DNA:DNA homoduplex (Giesen et al., 1998). PNAs have been mainly used together with ISH techniques for the detection of repetitive sequences for diagnostics purposes, such as centromeric and telomeric sequences, as well as ribosomal RNA (Stender et al., 2014; Pellestor et al., 2008; Stender, 2003; Pellestor et al., 2005). Moreover, the robustness using PNA probes has allowed to establish high-throughput quantitative methods for a precise analysis of the detected targets (Slijepcevic, 2001; Canela et al., 2007). However, the use of PNAs is restricted to the location of their targets, or a limited discrimination among sequences by using a set of probes (Gaylord et al., 2005). Then, the use of PNAs does not allow to specifically identify single bases contained in the detected sequences.

One of the repetitive sequences detected by using PNA probes is the α -satellite DNA. The α -satellite DNA constitutes the main component of centromeres and is organized in long arrays of tandem DNA repeats (Miga et al., 2014; Aldrup-MacDonald et al., 2016). These sequences represent the ~3% of the human genome and plays an important role in genomic stability (Miga and Black, 2017). The α -satellite DNA is organized in multimeric High Order Repeats (HOR), each one formed for the repetition of monomers of 171 bp-length (Shannon M McNulty, 2018). The biological implications of genomic variation in the sequence of the α -satellite DNA and other repetitive sequences remain unclear. Different studies point that variability in the α -satellite DNA, such as Single-Nucleotide Variations (SNVs), could have a major impact on chromosomal functions and cell division (Miga et al., 2014; Aldrup-MacDonald et al., 2016; Barra and Fachinetti, 2018). However, the study of these variability is limited. The reasons are that repetitive sequences have been poorly studied and excluded from genomic assembly studies because they are not protein-coding sequences (Miga et al., 2014; Miga and Black, 2017) and, mainly, because over the past decade, Next-Generation Sequencing methods have presented ambiguities in the assembly and interpretation of single-base differences in the α -satellite DNA and other repetitive sequences (Miga and Alexandrov, 2021; Tørresen et al., 2019; Cechova, 2021). Recently, a comprehensive study achieved a fully sequenced human genome, including satellite repeats (Nurk, 2022). This great achievement will allow a better understanding of the functions of these repetitive sequences. However, so far, there is a lack of techniques for the detection of the α -satellite DNA with single-base resolution in a direct and rapid manner. Such techniques could be a powerful tool for the validation of predicted SNVs by sequencing data and computational analysis, as well as for the detection of novel SNVs in repetitive sequences. These limitations urge to continue advancing for efficient experimental validation methods of sequencing data and to improve the resolution capabilities of ISH techniques, to achieve a better accuracy as well as to know details about the sequence of the detected nucleic acids.

To address the limitation of detecting SNVs in repetitive sequences, in this work, we report chemFISH, an innovative method for the detection of the human α -satellite DNA with single-base resolution by microscopy. ChemFISH is a method based on dynamic chemistry labelling and abasic PNA probes (Bowler et al., 2010, 2011), and, with this approach, we detected for the first time the α -satellite DNA *in situ*, as well as one single base contained in its sequence (referred as single-base resolution), in a direct and rapid reaction. The abasic PNA probes are chemically modified so that, in its strand, there is a nucleotide-free position (known as abasic position) that lies opposite to a single base under study in the nucleic acid target. When the abasic PNA hybridizes to its nucleic acid target, a chemical pocket is formed with a secondary amine group opposite to that single base under study. Then, aldehyde-modified nucleobases (called SMART-Nucleobases) (Bradley

and Díaz-Mochón, 2014) can be incorporated in this abasic position. Following the Watson and Crick pairing rules, only the complementary SMART-Nucleobase to the single base under study will react in the chemical pocket, generating an iminium intermediate. Then, a reduction step lock-up the SMART-Nucleobase as a stable tertiary amine in the complex. These SMART-Nucleobases can be labelled with biomolecules or fluorophores, thus detecting the specific SMART-Nucleobase incorporated to the chemical pocket and identifying the single base under study. This approach based on dynamic chemistry labelling has been previously implemented for the efficient detection of nucleic acids in liquid biopsy by mass spectroscopy, flow cytometry, electrochemical and colorimetric analysis (Angélica Luque-González et al., 2018; Marín-Romero et al., 2018; Delgado-Gonzalez et al., 2019; Tabraue-Chávez et al., 2019; Robles-Remacho et al., 2021).

Using this approach, we detected the α -satellite DNA *in situ* with single-base specificity, and, to assess the analytical performance, we explored the specific incorporation of the SMART-Nucleobase, the reproducibility in different human cell lines, and we performed a statistical analysis to know the consistency and robustness of the reported method. The method reported here could be used as a tool to detect novel SNVs in the human α -satellite DNA, as well as to validate SNVs predicted by computational analysis. In addition, the chemistry developed in chemFISH is versatile so that the abasic PNAs and the SMART-Nucleobases can be labelled with a range of different labels and strategies (Angélica Luque-González et al., 2018; Marín-Romero et al., 2018; Delgado-Gonzalez et al., 2019; Tabraue-Chávez et al., 2019; Robles-Remacho et al., 2021).

2. Materials and methods

2.1. Design and synthesis of the abasic PNA probe

The new PNA probe with the abasic position was designed 18-mer length and complementary to a sequence present in the α -satellite DNA of all the centromeric regions (Giunta and Funabiki, 2017). The PNA probe was designed labelled with Cy3-fluorophore and with the abasic position in the central region, represented as * (N-Cy3-xx-AACTAGA*AGAAGCATT-C), where “xx” represents diethylene glycol (miniPEG) as a spacer (for full details about the chemical structure and sequence of the abasic PNA probe, see [Supporting Material 1](#)). The synthesis was carried out based on standard solid phase synthesis techniques on polymeric supports (Tentagel resin (Polymer, United Kingdom)) using an Intavis Bioanalytical MultiPrep CF synthesizer (Intavis AG GmbH, Germany). The synthesis was performed by repeated rounds of coupling of activated PNA monomers with the amino group protected, followed by deprotection of the terminal amino group with washing steps after each round. The reactions were carried out at room temperature in microscale columns with a polytetrafluoroethylene filter (Intavis, Germany). The abasic PNA probe and SMART-Nucleobases were synthesized and characterized by DESTINA Genómica S.L. (Spain).

2.2. Cell cultures

All cell lines were provided by the Cell Bank of the Centre of Scientific Instrumentation of Granada. The cell lines HT-29 (colorectal adenocarcinoma) (ECACC 91072201), MDA-MB-468 (mammary gland adenocarcinoma) (ATCC HTB-132), HeLa (cervical adenocarcinoma) (ECACC 93021013) and MEF (*Mus musculus*) (fibroblasts) (ATCC SCRC-1040) were grown in DMEM medium (Gibco, Paisley, UK). The cell line H1975 (lung adenocarcinoma) (ATCC CRL-5908) was grown in RPMI medium (Gibco, Paisley, UK). Both media were supplemented with 10% fetal bovine serum (Gibco, Paisley, UK), 100 U/mL penicillin/streptomycin (Gibco, Paisley, UK), 1 × L-glutamine (Gibco, Paisley, UK) and 1 mM sodium pyruvate (Sigma Aldrich). The cell lines were grown at 37 °C in a 5% CO₂ humid incubator.

2.3. Hypotonic shock

Cell lines were grown in T25 flasks to 80% confluence and trypsinized ($1 \times$ trypsin-EDTA solution, Sigma Aldrich) at 37 °C for 5 min. Then, cells were collected and centrifuged for 5 min at 1500 rpm. The cell pellet was resuspended in 8 mL of hypotonic solution (0.075 M potassium chloride) and incubated at 37 °C for 30 min to obtain isolated nuclei. After this time, the nuclei were centrifuged for 5 min at 1500 rpm, the supernatant was removed and the fixative solution was added.

2.4. Fixation

The fixation was performed by slowly resuspending the cell pellet with a fixative solution consisting in 1 part of glacial acetic acid to 3 parts of methanol. Then, the sample was incubated for 30 min at 4 °C. The process was repeated 4 times and then the nuclei were resuspended in 1 mL of fixative solution. Finally, two drops of the suspension were added to a slide and airdried overnight. Up to 50 slides can be prepared at once per each ~80% confluent T25 flask. The slides that were not immediately used for FISH, they were stored at 4 °C. When needed, the slides were blocked for 1 h with 3% goat serum prior to the chemFISH reaction.

2.5. ChemFISH reaction

Before hybridization, slides were immersed twice in PBS for 2 min and then the samples were dehydrated by immersing the slides in increasing ethanol series (70% - 85% - 100%). Next, a chamber (Grace Bio-Labs, 9 mm diameter, 0.8 mm deep) was strongly fixed to the slide, filled with PBS, and sealed. Next, denaturation was carried out at 94 °C for 10 min in a Thermobrite (Abbott). After the denaturation, the slides were placed in ice for 2 min. Then, the slides were removed from the ice and the PBS was removed from the chamber. Next, the chamber was filled with the hybridization solution. The hybridization solution consisted in 10 mM phosphate buffer with pH carefully adjusted to 6, the abasic PNA probe at a final concentration of 50 nM, the SMART-Cytosine-REX-PEG₁₂-Biotin (henceforth referred to as SMART-C-Biotin) in a concentration of 5 μ M and the reducing agent (NaBH₃CN) at a final concentration of 1 mM and a final volume of 50 μ L. Two control samples were running in the same conditions, one without the SMART-C-biotin, and the other without the abasic PNA probe. Once the hybridization solution was added, the chamber was sealed and placed in a Thermobrite at 40 °C for 2 h. After the incubation, the chamber was removed, and the slides washed by immersion in $2 \times$ SSC 0.2% Tween-20 or 5 min at 37 °C. Then, a second wash was performed by immersion in $2 \times$ SSC 0.1% Tween-20 for 5 min followed by a water rinse, ready to proceed with the Tyramide Signal Amplification (TSA).

2.6. Tyramide Signal Amplification (TSA)

The TSA reaction was performed following the manufacturer's instructions with slight modifications (Tyramide SuperBoost™ kit with streptavidin and Alexa Fluor-488, ThermoFisher Scientific). Briefly, three washes in PBS were performed for 10 min each, followed by 1 h incubation with HRP-Peroxidase at room temperature in a humid chamber. Then, another three washes in PBS were performed for 10 min each. Next, the sample was incubated with 100 μ L of working solution according to manufacturer's instructions. The working solution contains the HRP-Peroxidase substrate (tyramide conjugated with Alexa Fluor 488) and H₂O₂. The reaction was incubated for 10 min. Next, the samples were rinsed three times with PBS and nuclear staining was carried out, depositing 5 μ L of mounting medium with DAPI (1.5 μ g/mL) (Vectashield Antifade). Then, the sample was covered with a coverslip and sealed. Slides were ready to be detected by confocal microscopy.

2.7. Confocal microscopy

Images used for quantification were obtained with a Zeiss LSM 710 inverted confocal laser microscope with a $63 \times$ /1.4, 1.0, 2.0 plan-apochromatic oil zoom factor. The lasers used were 405 nm diode laser at 3.0% for DAPI, argon laser 488 nm at 2.2% for Alexa Fluor 488 and HeNe laser 543 nm at 10.0% for Cy3. The acquisition was performed sequentially, maintaining each filter individually configured to avoid interference between channels. DAPI detection range: 410–487 nm; Alexa Fluor 488: 488–524 nm; Cy3: 543–618 nm.

2.8. Specific incorporation of SMART-C-Biotin

To determine the specific incorporation of SMART-C-Biotin, four independent assays were carried out, each one with one of the four biotinylated SMART-Nucleobases at 5 μ M. The SMART-Nucleobases used were SMART-C-biotin, SMART-Adenine-deaza-enol-PEG₁₂-biotin (henceforth referred to as SMART-A-biotin), SMART-Guanine-deaza-enol-PEG₁₂-biotin (henceforth referred to as SMART-G-biotin) and SMART-Thymine-REX-PEG₁₂-biotin (SMART-T-Biotin). Control samples were run in the same conditions, but without the biotinylated SMART-Nucleobase in each case. Reactions were performed in the same conditions above detailed (section 2.5), followed by the TSA (section 2.6) and visualized by confocal microscopy (as described in section 2.7).

2.9. Reproducibility and specific detection of human α -satellite DNA

The method was initially developed in the human cell line HT-29, which is a colorectal adenocarcinoma cell line and subsequently validated in the human cell lines MDA-MB-468 (mammary gland adenocarcinoma), H1975 (lung adenocarcinoma) and HeLa (cervical adenocarcinoma). To test the specific detection of α -satellite DNA of human species, a test was carried out by using a mouse (*Mus musculus*) cell line of fibroblasts, MEF. All reactions were performed in the same conditions above detailed (section 2.5 and 2.6) and visualized by confocal microscopy (as described in section 2.7).

2.10. Analytical performance and statistical analysis

To determine the analytical performance of the reported method, a quantification was carried out for both discrete signals: Cy3 (α -satellite DNA) and AF488 (single-base detection) signals. For this purpose, the assay was repeated three times in three cell lines, and then, 25 metaphase nuclei and 50 interphase nuclei were randomly imaged. The quantification was performed using ImageJ software (NIH, USA) and the number of discrete signals for both Cy3 and AF488 was averaged using GraphPad Prism software. The average number of red and green signals were compared to the modal number (number of chromosomes per nucleus) provided by ATCC (Manassas, VA, USA) of each cell line. Coefficients of variation for each cell line were calculated. Then, to know the sensitivity of single-base resolution, we compared both averaged discrete signals in a ratio AF488: Cy3.

3. Results and discussion

3.1. Detection of centromeric α -satellite DNA with single-base resolution by chemFISH

To assess the use of chemFISH, we carried out a proof of principle to detect the α -satellite DNA *in situ* with single-base specificity in the human colorectal adenocarcinoma cell line HT-29. We designed an abasic PNA probe complementary to a consensus sequence present in the 171-bp monomer of the human α -satellite DNA of all centromeres (Giunta and Funabiki, 2017). This PNA probe has the abasic position in the central region, represented as $^*_{-}$, and Cy3-fluorophore labelling at the N-terminus (N-Cy3-xx- AAAGTAGA $^*_{-}$ AGAAGCATT-C). Therefore,

due to the repetitive nature of centromeres, the abasic PNA probe hybridizes in tandem staining all centromeres in red (Cy3). Considering the consensus sequence of the α -satellite DNA, the abasic position lied opposite to a guanine in each monomer, so we used a biotinylated aldehyde modified Cytosine (SMART-C-biotin) to detect that guanine. After the hybridization of the abasic PNA probe, the SMART-C-biotin was incorporated to each abasic site and chemically lock-up by reductive amination. After the reaction, we revealed the biotin with Tyramide Signal Amplification (TSA), which deposits Alexa Fluor-488 fluorophores surrounding the reaction. Therefore, the single bases were detected as green signals (AF488). Under the microscope, the co-localization of both signals indicates the detection of the α -satellite DNA (Cy3) with single-base resolution (AF488) (Fig. 1).

Obtained results showed that both signals co-localized in the centromere of chromosomes in metaphase nuclei (Fig. 2). However, in interphase nuclei both co-localized signals were dispersed, due to the relaxed nature of chromatin. Notably, when the sample were running without the SMART-C-biotin as negative control, no signals were found in AF488 channel. In addition, when the sample was running without the abasic PNA probe, no signals were found in Cy3 or AF488 channels. Our results indicated that we detected successfully a sequence contained in the 171-bp monomer of the α -satellite DNA and a guanine in a specific position in this sequence. Remarkably, this method has been optimised to be done in less than 6 h using standard materials and reagents for microscopy analysis. Moreover, we benchmarked chemFISH with two commercial probes for PNA-FISH, showing a similar detection of the α -satellite DNA using both methods, confirming the successful detection of the α -satellite DNA with the abasic PNA probe, and, with the advantage that chemFISH allows the spatial detection of a single base contained in the sequence of the α -satellite DNA. All these data are

presented in [Supporting Material 2](#).

3.2. Specific incorporation of SMART-C-Biotin

To determine the specific detection of guanine, we carried out four independent assays using the abasic PNA probe under the same conditions but shifting the SMART-Nucleobase used in each assay. For each assay we used a different biotinylated SMART-Nucleobase (SMART-C-biotin, SMART-A-biotin, SMART-G-biotin and SMART-T-biotin). The results showed green signals (AF488) only when the SMART-C-biotin was used, so indicating the detection of a guanine (Fig. 3).

This result prove that a guanine is the predominant nucleobase in that position reinforcing the single-base specificity of this method. As we did not find green signals using other biotinylated SMART-Nucleobases, we concluded that there were no high-frequency polymorphisms in this position. With this approach, we detected the α -satellite DNA and a single base contained in its sequence by using a unique abasic PNA probe, in contrast with the approach of using a set of PNA probes, which is limited to thermal hybridization differences among the probes and does not detect specifically a single base (Gaylord et al., 2005; Chen et al., 1999).

Cross-reactivity between SMART-Nucleobases has been fully tested in previous studies (Bowler et al., 2011; Tabraue-Chávez et al., 2019; López-Longarela et al., 2020). With the aim to test the cross-reactivity between the four biotinylated SMART-Nucleobases used in chemFISH, we carried out a mass spectrometry analysis by using MALDI-TOF. The mass spectra showed that only the complementary SMART-Nucleobase can be incorporated to the abasic site, thus, detecting specifically the nucleobase under study. These data showed the specificity of the method lacking cross-reactivity between the biotinylated SMART-Nucleobases

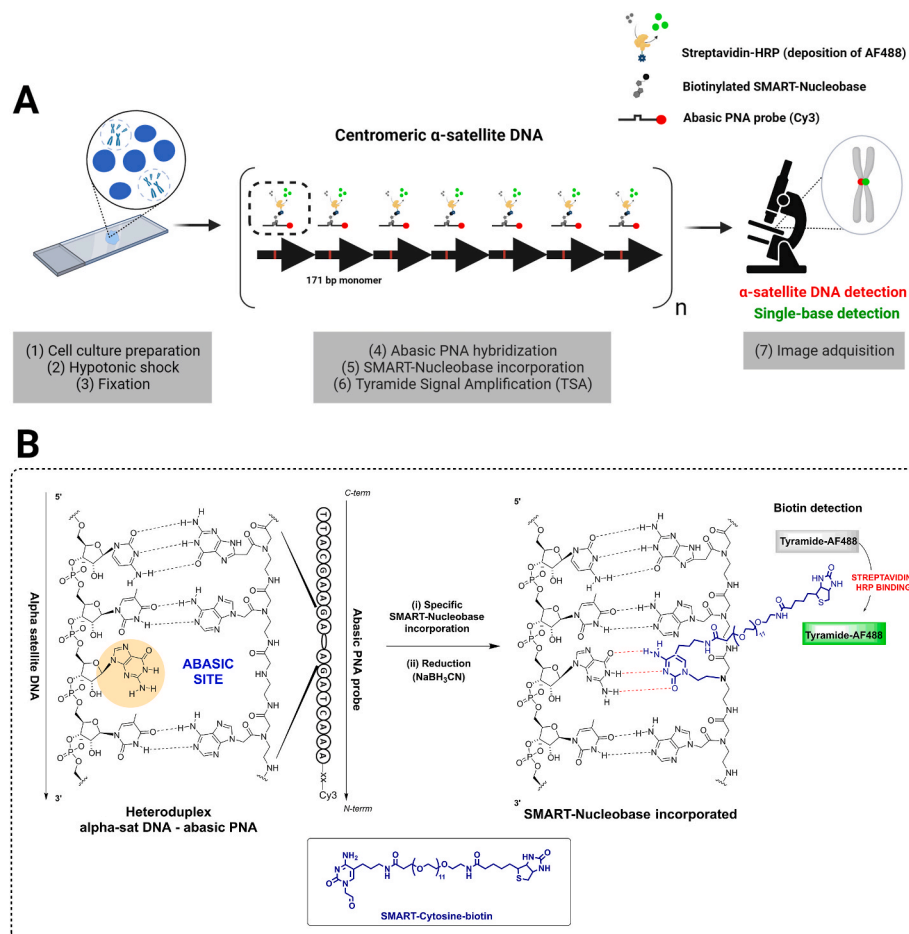


Fig. 1. Schematics of chemFISH reaction. A- Schematics of full analytical protocol of chemFISH. Cell culture was grown, trypsinized and incubated with a hypotonic solution to obtain isolated nuclei. The isolated nuclei were fixed and deposited in slides. Then, chemFISH was carried out incubating the isolated nuclei with the abasic PNA probe (Cy3-labelled) and the SMART-Nucleobase (biotin-labelled). After the reaction, the biotin was revealed with Tyramide Signal Amplification (TSA), what deposits tyramide-Alexa Fluor 488 surroundings the reaction. Under the microscope, red signals (Cy3) were indicative of the α -satellite DNA detection while green signals (AF488) were indicative of single-base detection. B- Chemical details of chemFISH reaction. The abasic PNA probe (Cy3-labelled) hybridized a sequence in the α -satellite DNA and the abasic position lied in front of a single base under study (highlighted in red). Then, the complementary SMART-Nucleobase (biotin-labelled) was incorporated in the abasic position. The reaction was lock-up by a reductive amination and biotin revealed with TSA.

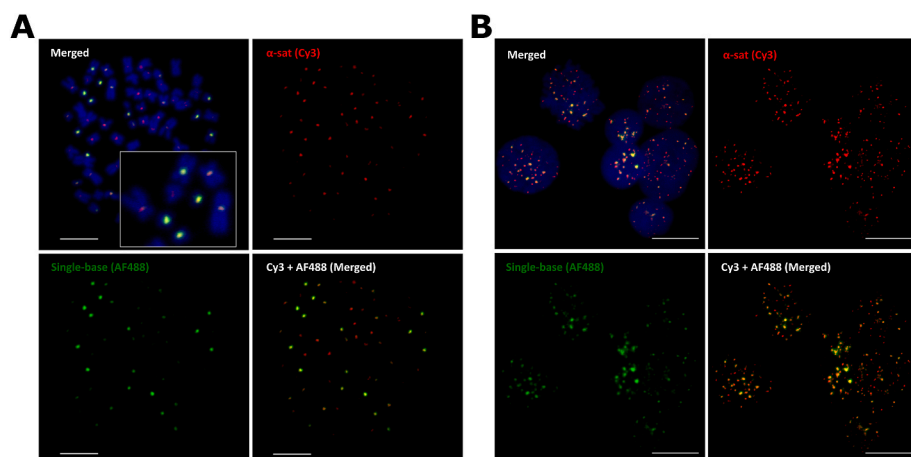


Fig. 2. ChemFISH for the detection of the α -satellite DNA (Cy3) with single-base resolution (AF488) in the centromere of metaphase (A) and interphase nuclei (B). A- Cy3 and AF488 discrete signals co-localized in the centromere of metaphase nuclei. B- Cy3 and AF488 discrete signals co-localized dispersed in interphase nuclei. Cell line: HT-29. Images obtained by inverted confocal microscopy (Zeiss LSM 710). $63\times$ magnification. Zoom factor (A): 2.0. Zoom factor (B): 1.5. Scale bar (A): $10\text{ }\mu\text{m}$. Scale bar (B): $20\text{ }\mu\text{m}$.

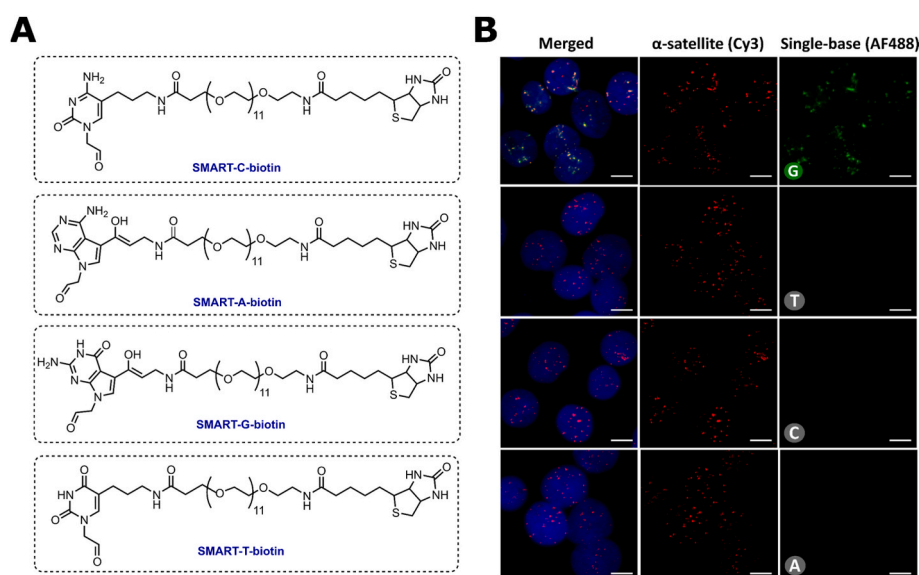


Fig. 3. Specific incorporation of SMART-C-biotin, detecting a guanine (AF488). A- Four different biotinylated SMART-Nucleobases were tested. B- Only when SMART-C-Biotin was used, there was signals in the AF488 channel, therefore, detecting a guanine in the α -satellite DNA. Cell line: HT-29. Images obtained by inverted confocal microscopy (Zeiss LSM 710). $63\times$ magnification. Zoom factor 1.5. Scale bar: $10\text{ }\mu\text{m}$.

used in chemFISH. All these data can be found in [Supporting Material 3](#).

3.3. Reproducibility and specific detection of human α -satellite DNA

In order to evaluate the versatility of this approach, we extended the method to the human cell lines MDA-MB-468 (mammary gland adenocarcinoma), H1975 (lung adenocarcinoma) and HeLa (cervical adenocarcinoma). We detected in all cases the human α -satellite DNA with single-base resolution, detecting guanine (Fig. 4). The abasic PNA probe used in our experiments is specific for human species, so, to determine the absence of non-specificity bindings to other centromeric regions of other species, we used a mouse (*Mus musculus*) cell line, MEF (fibroblasts). The results showed no detection of the α -satellite DNA or single-bases in the mouse cell line (Fig. 4). To see the full images obtained in each individual channel using different cell lines, see [Supporting Material 4](#). These results indicated that the method can be applied to different human cell lines. Also, these results indicated that the abasic PNA probe is specific of humans, as there were not non-specific signals of centromeric sequences of other species different to humans.

3.4. Analytical performance and statistical analysis

To assess the analytical performance of this method, we performed a statistical analysis. For that purpose, a quantification was carried out for both discrete signals: Cy3 (α -satellite DNA) and AF488 (guanine detection). The assay was repeated three times in the cell lines HT-29, HeLa and MDA-MB-468, and then, 25 metaphase nuclei were randomly imaged in each cell line. The quantification was performed using ImageJ software (NIH, USA) and the number of discrete signals for both Cy3 and AF488 was averaged as red (Cy3) or green (AF488) signals per nucleus. In order to know the sensitivity of single-base detection, we established the AF488: Cy3 ratio, that indicates the percentage of guanine detected in the position under study per each α -satellite DNA sequence detected. To establish the ratio, the percentage of guanine detected within the α -satellite DNA was calculated considering as 100% the number of hybridised spots detected by the Cy3 abasic PNA probe (red signal). Then, with the number of spots created by TSA signal (AF488) which are colocalised with the red signal, we calculated the percentage of guanine detected. We successfully detected a guanine in a range of 82%–89% of all detected α -satellite DNA. Also, we calculated the coefficients of variation for both signals. Coefficients of variation for red signals were

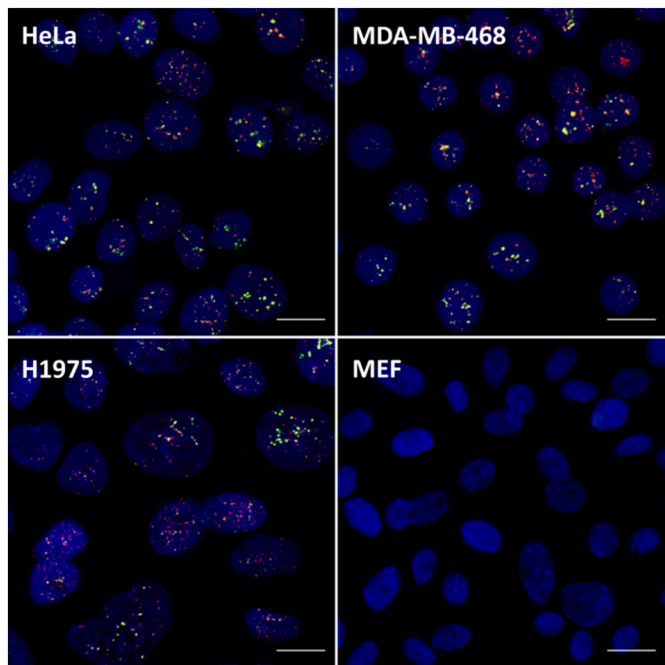


Fig. 4. ChemFISH reaction in different human cell lines and a mouse cell line. The human α -satellite DNA (Cy3 signals) with single-base resolution (AF488 signals) was detected in different human cell lines (HeLa, MDA-MB-468, H1975). There was no detection in the mouse cell line (MEF). All channel merged. Images obtained by inverted confocal microscopy (Zeiss LSM 710). 63 $\times$ magnification. Scale bar: 20 μ m.

24.80% (HT-29) 13.16% (HeLa), and 25.33% (MDA-MB-468). These coefficients of variations fall within the range obtained when using commercial FISH alternatives for the detection of the α -satellite DNA (explained by the aberrant modal number of the HT-29 tumour cell line) (Supporting Material 2). Coefficients of variation for green signals were 20.95% (HT-29), 16.29% (HeLa), and 24.13% (MDA-MB-468). These data showed a low variability detecting the α -satellite DNA with single-base resolution among samples.

As the method stains all centromeres, we expected to detect one discrete Cy3 and AF488 signal per centromere per chromosome. Thus, we compared the average number of Cy3 and AF488 discrete signals per nucleus with the modal number (number of chromosomes per nucleus) provided by different references (ECACC General Cell Collection, 2022; ECACC General Cell Collection, 2022; American Type Culture Collection (ATCC), 2022). We detected the average of Cy3 discrete signals (α -satellite DNA) in a consistent range compared with the modal number of other references as well as the signal detected using other established commercial methods (Supporting Material 2). Consistently with the sensitivity for guanine detection, the average of AF488 discrete signals was in a lower range to the modal number in all cell lines (Fig. 5).

Individual data of the average of discrete signals, coefficients of variation and performance ratio are represented in Table 1.

In addition, we performed a statistical analysis in interphase nuclei. For this end, the assay was repeated three times for each cell line, and then, we increased the number of interphase nuclei randomly imaged to 50 in each cell line. The reason for that increase is that in interphase nuclei the chromatin is relaxed, and single centromeres can be difficult to identify, due to different centromeric regions being near each other and generating one single discrete signal. Consistently with this reasoning, we found a lower average number of Cy3 and AF488 discrete signals for each cell line compared to the modal number and the coefficients of variation was high among samples. The full statistical analysis of the average of discrete signals, coefficients of variation and performance ratio for each discrete signal in interphase nuclei can be found in Supporting Material 5.

The statistical analysis showed that chemFISH is consistent analysing discrete signals in metaphase nuclei, with reproducibility among samples and different human cell lines. We found low variability when analysing metaphase nuclei, and we successfully detected single bases in the 85.52% of the detected α -satellite DNA sequences. This result remark the novelty of the sensitive detection of the α -satellite DNA with single-base resolution in a direct and rapid assay while maintaining its spatial location.

Table 1
Statistical analysis of the quantification of the signal obtained from the α -satellite DNA (Cy3) detection and the single-base (AF488) detection in 25 metaphase nuclei of different cell lines. The average of discrete Cy3 or AF488 signals is represented as mean $\pm$ SD.

Cell line (25 metaphase nuclei)	HT-29	HeLa	MDA-MB-468
Average of Cy3 discrete signals (red)	70.96 $\pm$ 17.60	61.92 $\pm$ 8.149	64.72 $\pm$ 16.39
Coefficient of variation of Cy3 discrete signals (red)	24.80%	13.16%	25.33%
Average of AF488 discrete signals (green)	58.48 $\pm$ 12.25	55.00 $\pm$ 8.958	55.24 $\pm$ 13.33
Coefficient of variation of AF488 discrete signals	20.95%	16.29%	24.13%
Performance: Ratio AF488: Cy3	82.41%	88.82%	85.35%
Modal number according to references	68-72 ()	Aneuploid (ECACC General Cell Collection, 2022)	60-67 (American Type Culture Collection (ATCC), 2022)

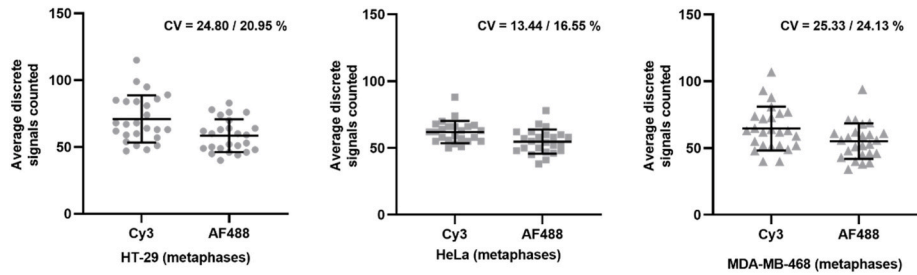


Fig. 5. Comparison of the average number of Cy3 discrete signals (α -satellite DNA) and AF488 discrete signals (single-base detection) detected per nucleus in 25 metaphase nuclei of different cell lines. Coefficient of variation (CV) is represented in the graphics as CV = % CV Cy3 discrete signals/% CV AF488 discrete signals.

4. Conclusions

Here, we report the development of chemFISH, an innovative method to detect *in situ* the α -satellite DNA with single-base resolution in a direct and rapid reaction. ChemFISH is a method based on dynamic chemistry labelling and abasic PNA probes. We detected by microscopy the α -satellite DNA with single-resolution in different human cell lines, showing reproducibility in different human cell lines and specificity to human species. We analysed the absence of cross-reactivity between the four biotinylated SMART-Nucleobases, showing specificity. We tested the analytical performance, finding a low variability (coefficients of variation ranging from 13.16% to 25.33%) and we detected single bases with high sensitivity (we detected single-base specificity in a range from 82.41% to 88.82% of all detected α -satellite DNA). The method and the analysis can be carried out in less than 6 h, using standard materials and reagents for microscopy analysis.

ChemFISH is a promising method that can serve to validate previously predicted SNVs by sequencing data and computational analysis, as well as to find novel SNVs in repetitive sequences. The molecular development is compatible to detect other repetitive sequences typically detected by using PNAs and ISH techniques, such as telomeric sequences and ribosomal RNAs (Stender et al., 2014; Pellestor et al., 2008; Stender, 2003; Pellestor et al., 2005). The method is limited to the detection of abundant nucleic acids, however, as future approaches, the method could be compatible to detect transcripts as well, as there are ISH techniques based in an isothermal amplification to generate artificial repetitive sequences around the transcripts, facilitating their detection (Ke et al., 2013; Lee et al., 2015). In addition, the chemistry behind chemFISH is versatile as the abasic PNA probe as well as the SMART-Nucleobases could be labelled and conjugated to different type of biomolecules and fluorophores, as previous studies have shown (Angélica Luque-González et al., 2018; Marín-Romero et al., 2018; Delgado-Gonzalez et al., 2019; Tabraue-Chávez et al., 2019; Robles-Remacho et al., 2021). In conclusion, the method presented here could contribute to the development of novel molecular assays based on PNA and ISH techniques to detect abundant nucleic acids *in situ*, as well as information contained in their sequences.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Juan Jose Diaz-Mochon reports a relationship with Destina Genomica SL that includes: equity or stocks. Salvatore Pernagallo reports a relationship with Destina Genomica SL that includes: employment and equity or stocks. Francisco Javier Lopez-Delgado reports a relationship with Destina Genomica SL that includes: employment. Juan Jose Guardia-Monteagudo reports a relationship with Destina Genomica SL that includes: employment. Mario Antonio Fara reports a relationship with Destina Genomica SL that includes: employment.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bios.2022.114770>.

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